

1a. Introduction to R

ECO1400

1. R and RStudio

R is a freely available software environment for statistical computing and graphics. RStudio provides a user-friendly interface to R. You can choose to use R online (typically useful for Chromebooks or tablets) or install and use R on your device (typically useful for laptops or notebooks).

1.1 Online Use

Follow the instructions under this [link](#) to run R and RStudio in a browser online in the cloud without installing it on your device. You will be asked to sign up for an account. Once signed in, create a New Project. Click on Upload (bottom right pane) to import script and data files into your cloud directory, the Open file (top left pane) to start using it.

Note that there is a (generous) time limit to access your cloud account for free. The time is based not on computing time used but rather login time. Remember to log out when you finish working with R online.

1.2 Installation on a Device

Follow the instructions under this [link](#) to install R (including Rtools) and RStudio on your device. This option provides an unlimited free access to R.

2. R Packages

Packages are collections of functions that can be used in your R code. More than 10,000 well-documented packages are freely available; this is one of the core strengths of R. When RStudio opens it automatically loads the package `base` containing some common functions. If we want to use additional functions we need to install and load the packages that contain them.

Specify the source of the packages:

```
options(repos = c(CRAN="http://cran.rstudio.com"))
```

We don't need R to install the requested packages every time the .Rmd file is run, so we specify that the given packages should be installed only if they are not already installed. For example, to install the package `stats`:

```
if (!require("stats")) install.packages("stats") # Statistical functions ACF and PACF
```

Once the package is installed it also needs to be loaded into R so that its functions are available to us. An installed package is loaded in by the command `library()`.

```
library(stats)
```

The R [help files](#) contain the description and a detailed list of functions of each package.

3. Markdown

R [Markdown](#) is a combination of R code, output, and comments, in a unified authoring framework for data science. R Markdown documents support many [output formats](#), such as PDFs (with LaTeX), Word files, html pages, slideshows, and more.

R Markdown files are text files saved with the .Rmd extension. They contain:

- A header with the title and output options (here pdf_document),
- Chunks of R code,
- Text with simple text formatting like **bold**, *italics*, or math notation such as $\alpha = \beta$ or $Y = \sum_{n=1}^{10} n^2$. [LaTeX](#) syntax is supported.

A heading is created by including the symbol # (or ## for subheading) at the beginning of the line, as throughout this document.

3.1 Running R code with Markdown in RStudio

When you open an .Rmd file in RStudio, you get a notebook interface where code, output, and comments are interleaved.

R code in RStudio is organized in “chunks”, which are shaded spaces enclosed between {r} and).

Example of an R code chunk: assign value 1 to object x using the assignment operator <-, and display the result.

```
x <- 1  
x
```

```
## [1] 1
```

You can run R code in various segments, by using the RStudio **Run** menu, or by running:

- The entire code chunk by – clicking the “play” button (green triangle) at the top of the chunk – pressing Cmd/Ctrl + Shift + Enter
- A single code line in the chunk by pressing Cmd/Ctrl + Enter
- All code chunks in the file by pressing Ctrl+Alt+R (Cmd+Option+R).

3.2 Converting Markdown output to PDF documents

To produce a document containing all text, code, and results, click “Knit” in the RStudio menu above or press Cmd/Ctrl+Shift+K.

For creation of PDF documents we need to have a LaTeX distribution installed. It is recommended that R Markdown users install [TinyTeX](#), which is small in size and does not require computer administrative privileges. TinyTeX is installed and loaded as follows:

```
if (!require("tinytex")) install.packages("tinytex")  
library(tinytex)  
  
# This command needs to be run once for initializing tinytex  
tinytex::install_tinytex()  
# RStudio will then use the tinytex installation for all further knits
```

4. Loading and Saving Data

4.1 Loading from File

Read data on GDP of Canada from a file and store the dataset in object `table`. Here the data is stored in a `.csv` file and so we use the function `read.csv` for reading it. For other types of files we can use the more general `read.table()` function. The argument `header=TRUE` indicates that the first row of the data contains variables names.

```
table <- read.csv(file="GDP_Canada.csv", header=TRUE, sep=",")
```

Our data is located in the same folder (directory) as the R file that reads the data. If the data is in a different folder, we need to specify the full path to it in the `file=` argument, e.g. `file=C:/Documents/EC0374/data/GDP_Canada.csv`.

We can check the current directory with:

```
wd <- getwd()
wd
```

```
## [1] "/Users/apple/Desktop/W26 375/Tut 1"
```

We can change the current directory with:

```
setwd(wd)
```

4.2 Loading data from an Online Source

There are many ways of reading data directly from an online source into R. A useful function for this purpose is `getSymbols()` in the package `quantmod`.

```
if (!require("quantmod")) install.packages("quantmod")
library(quantmod)
```

Let's read in financial data on Google (Alphabet) stock price ([Source](#)) from Yahoo finance.

```
GOOG <- quantmod::getSymbols("GOOG", src="yahoo", return.class="xts", auto.assign=FALSE)
head(GOOG) # display first few lines of the `data` file
```

```
##           GOOG.Open GOOG.High GOOG.Low GOOG.Close GOOG.Volume GOOG.Adjusted
## 2007-01-03  11.60650  11.87200  11.48470   11.64610   309415434    11.55940
## 2007-01-04  11.68122  12.05357  11.66503   12.03638   316686586    11.94678
## 2007-01-05  12.01746  12.14199  11.90812   12.13427  275914333    12.04394
## 2007-01-08  12.14672  12.20102  12.00998   12.04436   190888826    11.95469
## 2007-01-09  12.09093  12.16067  11.98508   12.09218   216062832    12.00216
## 2007-01-10  12.06553  12.29267  12.00600   12.19081   239634856    12.10005
```

5. R Online Resources

- R Cookbook, 2nd Ed ([link](#))
- Hands-On Programming with R ([link](#))